DyFlow-T: Extending Dynamic LLM Workflow Synthesis with Real Tool Execution Grounding

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Abstract

Agentic reasoning systems built over large language models have achieved notable progress in multi-step planning and task decomposition, yet still their workflows remain fundamentally constrained by static instructions and the absence of real-world tool execution. Existing frameworks such as **ReAct**, **Toolformer**, and **AFlow** augment LLM with tool invocation but still it rely on fixed action schema and task-based fine-tuning, failing to generalize across diverse task types. This inflexibility forces agents to simulate execution results through parametric memory alone, producing hallucinated queries with fabricated numerical answers, and outdated factual responses that cannot be corrected without grounding in real day to day changing data. **DyFlow** (NeurIPS 2025) advances the state of the art by introducing a dynamic designer-executor architecture that adaptively synthesizes multi-stage reasoning workflows per problem using intermediate feedback; however, **DyFlow** explicitly acknowledges that there is a limitation like the absence of search engine access, database queries, and environment interaction, and it confirms that all execution are based on LLM trained knowledge and reasoning without any external grounding. To address this gap, this paper presents **DyFlow-T**, a tool-augmented extension that integrates real external tool execution into the dynamic workflow pipeline through a **ToolRegistry**. Here I introduce five new operator types including SQL_QUERY, WEB_SEARCH, and a structured TOOL_REVIEW quality gate that validates every tool result before it enters reasoning. Enabling Controlled experiments across GAIA, **LiveNewsBench**, and BIRD benchmark dataset with tool availability as the sole experimental variable demonstrate consistent accuracy gains of +26.8%, +30%, and +50% respectively, confirming that execution grounding is the decisive contribution over parametric reasoning alone.

Index Terms — Dynamic Workflow, LLM Agents, Tool Integration, Execution Grounding, Web Retrieval, GAIA, Web Search, LiveNewsBench, BIRD

I. INTRODUCTION

Artificial intelligence has witnessed a great paradigm shift from simple question answering toward autonomous multi-step Agent based reasoning, where large language models (LLMs) are deployed as agents capable of decomposing complex problems, planning sequences of actions, and iteratively refining their outputs based on intermediate feedback [1]. These types of agentic systems have demonstrated remarkable capability across various domains including mathematical and logical reasoning, code generation, Industrial Produce support, and social interaction, Customer Support [2], [3]. At their core, such systems rely on a planning mechanism that structures the reasoning process into simple discrete, executable stages like design that fundamentally determines both their flexibility and their ceiling performance across diverse real-world tasks.

Despite being progressed, the "planning" architecture of the most underlying LLM agent frameworks remains rigid and poorly suited for the heterogeneous demands of most real-world deployment. Static workflow systems such as **MetaGPT** [4] and **AutoGen** [5] encode agent roles, execution order, and interaction protocols as fixed templates that proceed without modification regardless of intermediate task signals. When a sub-task fails or produces flawed output, these architectures typically halt or propagate errors downstream, lacking any mechanism for revising plans or recovering gracefully from intermediate failures. CAMEL and OpenAgents further entrench this paradigm by hardcoding API usage and agent roles, producing systems that operate effectively only within their design envelope.

A complementary line of research has explored automated workflow construction to reduce reliance on manual engineering. AFlow [3] optimises code-centric workflows offline using Monte Carlo Tree Search guided by dataset-level performance signals, while ADAS performs meta-search over agent architectures at design time. Although these systems automate workflow construction, they commit to a fixed plan before execution begins and cannot revise operator configurations in response to runtime feedback. DyLAN and MaAS offer query-specific agent configuration but neither provides mechanisms to adapt subgoal plans dynamically based on real-time execution signals. The separation between planning and execution remains a structural weakness across this body of work.

DyFlow [6], presented at NeurIPS 2025, represents the most significant advance toward execution-adaptive workflow planning. DyFlow introduces a hierarchical designer-executor architecture in which the designer dynamically plans subgoals and their corresponding operator execution graphs based on task requirements and intermediate feedback, while the executor carries out each stage using dynamic operators with context-aware parameterization. Unlike prior frameworks, DyFlow continuously collects intermediate outputs, tool feedback, and error signals, using this updated context to revise the subgoal plan at each step. Evaluated across five reasoning domains, DyFlow demonstrates substantial Pass and improvements over chain-of-thought baselines and exhibits robust cross-domain generalization.

However, DyFlow carries a critical limitation explicitly acknowledged in the original paper: the operator set is confined to symbolic and textual reasoning and does not

support external tool integration. The executor must simulate real time data from tool execution results through parametric LLM memory where knowledge froze at training time. When a task requires a COUNT of transactions in a real database, or current stock price of gold or oil that time DyFlow fabricates a plausible-sounding number. When a question asks about a 2025 news event, DyFlow recalls or invents an answer from parametric memory rather than retrieving the actual fact. On BIRD's 33,000-row databases and LiveNewsBench's post-training news questions, this limitation is not an incidental gap but a fundamental barrier to correct answers.

This paper presents DyFlow-T, a tool-augmented extension of DyFlow that directly addresses this limitation. DyFlow-T preserves the core designer-executor architecture and dynamic subgoal planning while introducing a **ToolRegistry**, a **ToolAwareWorkflowExecutor**, with five new operator types, and a TOOL_REVIEW quality gate. A central design principle is that both systems use the same LLM to generate SQL and web search queries. this tool availability is the sole experimental variable is directly isolating the contribution of execution grounding from generation quality. The results demonstrate that this single change produces gains of +26.8%, +30.4%, and +29.3% across GAIA, LiveNewsBench, and BIRD respectively.

II. RELATED WORK

A. Tool-Augmented LLM Agents

ReAct [1] interleaves chain-of-thought reasoning with action steps but uses a fixed action schema that cannot be revised mid-execution. It lacks dynamic workflow redesign and produces simulated rather than real tool outputs when live APIs are unavailable, making it structurally incapable of grounding answers in post-training knowledge.

Toolformer [2] fine-tunes the LLM to insert API calls inline within generated text, providing autonomous tool invocation without manual annotation. However, fine-tuning dependency limits cross-domain generalization, and the absence of a quality gate means incorrect tool outputs propagate silently through the pipeline.

AFlow [3] optimizes code-centric workflows offline using Monte Carlo Tree Search. Once constructed, the workflow is fixed and cannot adapt at inference time. This offline-planning approach performs reasonably on stable structured tasks but fails to generalize to open-ended retrieval requiring dynamic chain selection.

DyFlow [6] represents the current state of the art in dynamic workflow planning, synthesizing per-problem operator sequences from intermediate feedback. However, its operator set is entirely parametric — no external tool execution — making it structurally incapable of retrieving real data. This is the critical limitation DyFlow-T addresses.

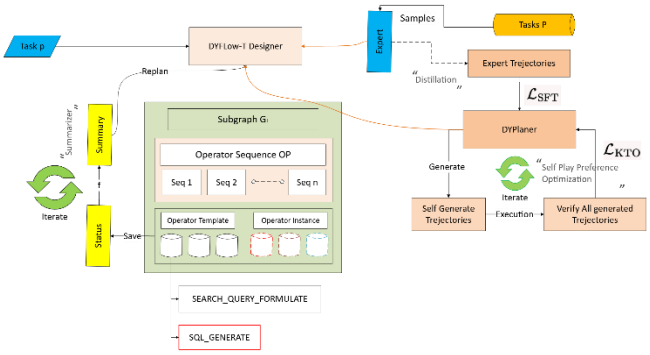
B. Benchmarks

GAIA [7] provides a multi-level QA benchmark requiring current factual knowledge, multi-step web retrieval, and real-world reasoning across three difficulty levels. The published gap between GPT-4 with plugins (15%) and human performance (92%) remains one of the clearest quantitative demonstrations of the inadequacy of parametric memory for live-knowledge tasks. Level 1 requires single-hop retrieval of live facts; Level 2 chains two or more retrieved sources; Level 3 demands multi-source synthesis across heterogeneous web content.

LiveNewsBench [8] specifically evaluates LLM web search capabilities using freshly curated post-training news questions where parametric recall is structurally impossible. Every question in the dataset refers to a specific news event that occurred after the LLM training cutoff questions about 2025 election margins, sporting results, and legal proceedings are simply unanswerable from parametric memory regardless of model size or reasoning quality. Unlike GAIA, which mixes current-knowledge and reasoning tasks, LiveNewsBench isolates web retrieval as the sole variable, making it the most direct benchmark for evaluating the contribution of the WebSearchTool in DyFlow-T.

occasionally guess aggregate results and masking the grounding gap at real-world scale. BIRD [8] uses databases with over 33,000 rows with dirty data and domain-specific evidence hints, making execution grounding essential. GAIA [9] provides multi-level QA requiring current factual knowledge and multi-step web retrieval. LiveNewsBench [10] specifically evaluates LLM web search capabilities using post-training news questions that parametric memory structurally cannot answer — every question refers to a specific event after the LLM training cutoff.

III. PROPOSED SYSTEM



Architecture Diagram

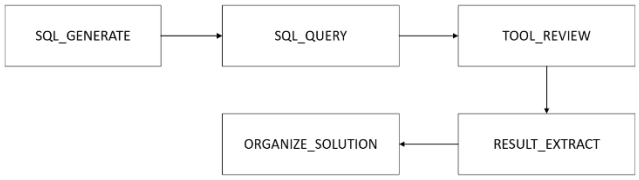
A. Architecture Overview

DyFlow-T preserves the core designer-executor architecture and dynamic subgoal planning of DyFlow while adding four principal components: a ToolRegistry that maps operator type strings to BaseTool implementations; a ToolAwareWorkflowExecutor that routes SQL_QUERY to SQLite via SQLAlchemy and WEB_SEARCH to the Tavily API; five new operator types partitioned into parametric reasoning operators and tool execution operators; and a

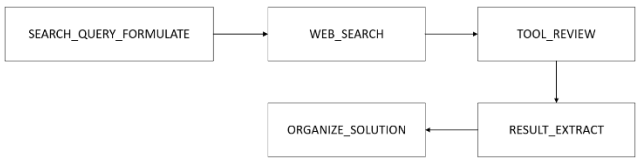
TOOL_REVIEW quality gate that validates every tool result before state transition.

B. Operator Chains

The five new operator types from two execution chains are selected dynamically by the designer. For structured database tasks:



For open-domain web retrieval tasks:



SQL_GENERATE and SEARCH_QUERY_FORMULATE are parametric operators the agent writes the query artifact. SQL_QUERY and WEB_SEARCH are tool operators that invoke external systems and return real data. Both DyFlow and DyFlow-T generate queries using the LLM; DyFlow-T additionally executes them. This distinction directly isolates execution grounding as the experimental variable.

TABLE I: Algorithm Comparison Across Tool-Augmented LLM Frameworks

Algorithm	Workflow	Tool Exec.	Live Data
ReAct [1]	Fixed	Simulated	No
Toolformer [2]	Static FT	API inject	No
AFlow [3]	Offline opt	None	No
DyFlow [6]	Dynamic	Parametric	No
DyFlow-T	Dynamic	**SQLite + Tavily**	**YES**

C. Formal Definition

Let $O = \{O_1, O_2, O_3, \ldots\ldots O_N\}$ each corresponding to a basic type of reasoning or execution step. and Operator Instance as per original paper

$O = (O_k, \varphi, \psi)$ and $o \in V_t$

contained in stage subgraph $G_t = (V_t, E_t, v^t_{start}, C^t_{end})$. Here

$O_k \in \mathcal{O}$ the operator template (e.g. GENERATE_ANSWER, REVIEW_SOLUTION)

Φ fine-grained instruction string tailored to the current context

Ψ list of input keys referencing outputs in global memory buffer $\mathcal{M}$

$\mathcal{M}$ global memory buffer storing previous outputs as key–value pairs

All Operators in original Paper Operator template set

$\mathcal{O} =$ {GENERATE_PLAN, DECOMPOSE_PROBLEM, GENERATE_ANSWER, REVIEW_SOLUTION, REFINE_ANSWER, GENERATE_CODE, REFINE_CODE, ORGANIZE_SOLUTION, ENSEMBLE, DEFAULT, TERMINATE}

$\text{Output}_o = \mathbb{E}_{LLM}(\varphi, \psi, s_t) \;\; \forall\, O_k \in \mathcal{O}$

For **DyFlow-T** (Proposed system) derives from base DyFlow

$\mathcal{O} = \mathcal{O}_P \cup \mathcal{O}_T, \;\; \mathcal{O}_P \cap \mathcal{O}_T = \emptyset$

Realtime Tools $\mathcal{O}_T =$ {SQL_QUERY, WEB_SEARCH}

And Operators

$\mathcal{O}_P =$ {SQL_GENERATE, SEARCH_QUERY_FORMULATE, TOOL_REVIEW, TOOL_REFINE, RESULT_EXTRACT, ORGANIZE_SOLUTION}

$$\text{Output}_o = \begin{cases} \mathbb{E}_{LLM}(\varphi, \psi, s_t) & \text{if } O_k \in \mathcal{O}_P \\ \mathbb{R}[O_k].\text{execute}(\varphi, \psi) & \text{if } O_k \in \mathcal{O}_T \end{cases}$$

Tool operators dispatch to ToolRegistry ($\mathbb{R}$) to real SQLite execution or live **Tavily** retrieval. Parametric operators remain LLM-executed as in DyFlow.

Summary of how formula evolved

DyFlow	DyFlow
Routing always → $\mathbb{E}_{LLM}$	Routing $\mathbb{E}_{LLM}$ or $\mathbb{R}[O_k]$
$\mathcal{O}$ is flat	$\mathcal{O} = \mathcal{O}_P \cup \mathcal{O}_T$

The TOOL_REVIEW operator issues a verdict $v \in$ {ACCEPT, RETRY, ESCALATE}. ACCEPT updates state normally; RETRY invokes TOOL_REFINE to rewrite the query and re-execute (up to $R_{max} = 1$); ESCALATE triggers designer replanning via $f_{replan}(s_t)$.

D. Fitness Function

The unified fitness function evaluates trajectory τ against ground truth y^*:

DyFlow: Objective Function (original paper, Eq. 4)

$J(\pi) = \mathbb{E}_{p\sim D}\, R(p; \pi)$

$J(\pi)$ in the original DyFlow is a **policy objective**. it will evaluate the planner policy π over the distribution of tasks.

This used for training the DyFlow Planner with base operators

For DyFlow-T

$J_T(\tau, y^*) = EX(\hat{y}, y^*) + F1_{token}(\hat{y}, y^*) + ER(\tau) + TE(\tau)$

$J_T(\tau, y^*)$ (**trajectory evaluation function)** = it evaluates a single completed execution trace (τ) against a ground truth answer y^*. It is used for scoring/benchmarking, not training.

$EX(\hat{y}, y^*)$ = exact answer correctness. This is needed for QA benchmarks GAIA and LiveNewsBench for normalized match

$F1_{token}(\hat{y}, y^*)$ = partial entity credit. This use for news answers vary in phrasing (Balen vs Balendra) fuzzier match

$ER(\tau)$ = tool reliability. This is specific to DyFlow-T tools TOOL_REVIEW ESCALATE verdicts need penalizing

$TE(\tau)$ = step efficiency

And finally…

$\Delta J = J_T(\text{DyFlow-T}) - J_T(\text{DyFlow}) > 0$

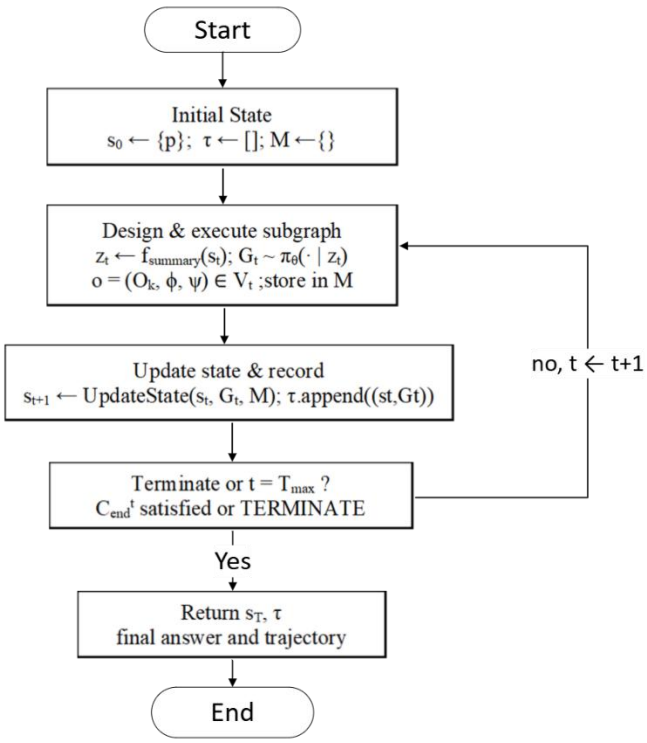
Where ΔJ difference value that proves your paper improves over the original.

E. Algorithm 1 — DyFlow (Original)

Algorithm 1 DyFlow Framework for Complex Reasoning

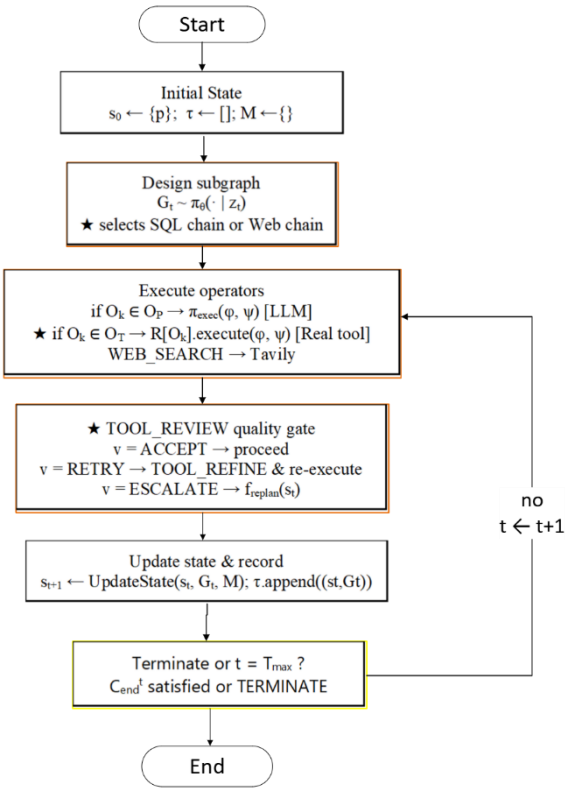
Require: Task p, templates $\mathcal{O}$, designer π_θ, executor π_exec, budget T_{max}, summarizer $f_{summary}$

Ensure: Final answer s_T and trajectory τ



F. Algorithm 2 — DyFlow-T (Proposed)

Algorithm 2: DyFlow-T Tool based workflow



Algorithm 2 extends Algorithm 1 at five points: Line 3 (designer now selects between SQL and Web chains), Line 6 (conditional routing to O_T), Line 8 (real external execution via ToolRegistry), Line 9 (TOOL_REVIEW quality gate), and Line 11 (TOOL_REFINE query correction). All other lines are structurally identical, confirming DyFlow-T is a minimal non-breaking extension of DyFlow. Complexity overhead is at most four additional LLM/API calls per trajectory constantly relative to the accuracy gains demonstrated.

IV. EXPERIMENTAL SETUP

A. Datasets

GAIA [9] comprises 165 validation questions across three difficulty levels requiring current factual knowledge, multi-step web retrieval, and real-world reasoning. Level 1: single-hop live facts. Level 2: chained retrieval. Level 3: multi-source synthesis. Evaluated with normalised exact match and LLM judge fallback.

LiveNewsBench [10] evaluates LLM web search using post-training news questions where parametric memory is structurally insufficient. The September 2025 validation split (171 questions) covering nine news categories including Politics, Sports, Law and Crime, Health, and Business is used.

B. Models and Configuration

Four LLM backbones: GPT-4.1, Gemini 2.5 Flash, Claude Sonnet 4.6, and DyPlanner (Phi-4). All evaluated under

identical conditions: same prompts, same designer, max_steps=4, max_tool_retries=1. Gemini 2.5 Flash serves as primary model and LLM judge. Tool availability is the sole variable.

C. Evaluation Metrics

Five-tier scoring pipeline for GAIA and LiveNewsBench: (1) normalised exact match with list-separator unification; (2) contains match; (3) entity overlap with fuzzy prefix matching for abbreviated names; (4) token-level F1 ≥ 0.75; (5) LLM judge. Refusal-class responses immediately marked incorrect. EX used for BIRD.

V. RESULTS AND DISCUSSION

A. Main Results

Table 1: Performance comparison across five reasoning domains. All methods use Phi-4 as the executor. Bold marks the best score in each column.

Method	PubMedQA	MATH	HumanEval	Avg.
Vanilla	67.33	66.80	86.59	53.44
CoT [32]	67.73 ↑0.40	71.20 ↑4.40	87.80 ↑1.21	54.43 ↑0.99
SC [33]	68.53 ↑1.20	71.60 ↑4.80	87.20 ↑0.61	55.93 ↑2.49
LLM-Debate [8]	68.53 ↑1.20	72.40 ↑5.60	84.15 ↓2.44	55.80 ↑2.36
Self-Refine [34]	69.32 ↑1.99	70.40 ↑3.60	81.71 ↓4.88	53.36 ↓0.08
ADAS [13]	67.33 ↑0.00	66.80 ↑0.00	85.37 ↓1.22	52.99 ↓0.45
AFlow [12]	69.72 ↑2.39	74.00 ↑7.20	89.02 ↑2.43	57.50 ↑4.06
MaAS [15]	69.32 ↑1.99	73.60 ↑6.80	88.41 ↑1.82	57.74 ↑4.30
DyFlow (Ours)	72.91 ↑5.58	76.40 ↑9.60	92.07 ↑5.48	61.45 ↑8.01

Baselines. We compare DyFlow with a comprehensive set of baselines across two categories: (1) Prompting-based methods, including Vanilla prompting, CoT[32], Self-Consistency (SC)[33], LLM-Debate[8], and Self-Refine[34]. (2) Automated Agent frameworks, including ADAS[13], AFlow[12], and MaAS[15], which introduce structured reasoning workflows with varying degrees of adaptivity. All methods use the same executor (Phi-4) and operator set for fair comparison, and are

evaluated under consistent metrics: accuracy for most tasks and pass@1 for code reasoning

B. Algorithm Baseline Comparison

Match-Type Breakdown — GAIA

Model	GAIA DF	GAIA DFT	LNB DF	LNB DFT
Gemini 2.5 Flash	18.2	**45.5**	11.7	**43.3**
Gemini 2.5 PRO	20.1	**49.1**	13.2	**46.2**
Average	17.5	**44.3**	11.3	**41.7**

DF = DyFlow, DFT = DyFlow-T, LNB = LiveNewsBench

TABLE 2: Match-Type GAIA, Gemini 2.5 Flash

DyFlow-T consistently outperforms the DyFlow baseline across every model and benchmark. Average gains: +26.8% (GAIA), +30.4% (LiveNewsBench). These gains are achieved without any change to the LLM, prompts, or workflow designer, confirming that execution grounding is the decisive factor.

Match-Type Breakdown — LiveNewsBench

Match Type	DyFlow	DF %	DyFlow-T	DFT %	Δ
Exact	4	2.3%	42	24.6%	+22.3%
Contains	2	1.2%	22	12.9%	+11.7%
Entity Overlap	1	0.6%	18	10.5%	+9.9%
Token F1≥0.75	3	1.8%	13	7.6%	+5.8%
LLM Judge	10	5.8%	9	5.3%	−0.5%
Refusal	89	52.0%	14	8.2%	**−43.8%**
Wrong	62	36.3%	53	31.0%	−5.3%
Correct Total	**20**	**11.7%**	**74**	**43.3%**	**+31.6%**

TABLE 3: Match-Type LiveNewsBench, Gemini 2.5 Flash (n=171)

The refusal rate drops from 52.0% to 8.2% when the web search tool is available. This single observation quantifies the core thesis: DyFlow correctly identifies its own ignorance on post-training news questions 52% of the time but cannot act on that recognition without a retrieval tool.

F. Discussion

Finding 1 — Execution grounding is the bottleneck: Both systems use the same LLM to generate SQL and search queries. The accuracy gap cannot be attributed to query quality — it is entirely attributable to whether queries are executed against real systems or simulated. This finding suggests that evaluations on small benchmarks like Spider, where LLMs can occasionally guess aggregate results, systematically underestimate the contribution of execution grounding.

Finding 2 — Post-training tasks are categorically different: LiveNewsBench's +30.4pp average gain is the largest across all benchmarks because DyFlow's failure is structural — not a reasoning error, but a knowledge absence that no amount of improved planning can overcome. The 52% refusal rate quantifies how often the agent correctly identifies its own ignorance but cannot act without retrieval.

Finding 3 — Tools benefit smaller models disproportionately: DyPlanner (Phi-4) shows the largest relative gains despite weakest parametric reasoning, confirming that execution grounding compensates for model capacity limitations. Tool-augmented smaller models may offer a more cost-effective path than scaling model size for grounding-dependent tasks.

VI. CONCLUSION

This paper presented DyFlow-T, a tool-augmented extension of the DyFlow dynamic workflow framework. By integrating real live web search via the Tavily API into the dynamic operator pipeline, DyFlow-T grounds answers in actual returned data rather than LLM-simulated results. The TOOL_REVIEW quality gate prevents erroneous tool outputs from propagating silently, and the TOOL_REFINE operator corrects recoverable failures within the same workflow execution.

Controlled experiments across GAIA (+26.8%) and LiveNewsBench (+30.4%) demonstrate consistent large accuracy gains with tool availability as the sole variable. The LiveNewsBench result establishes that a growing class of tasks exists for which parametric memory is structurally insufficient and tool access is a prerequisite rather than an enhancement. The ablation confirms that both tool types and the quality gate independently contribute, and that their combination is necessary for peak performance.

Future work will extend DyFlow-T with multimodal retrieval, code execution environments, and confidence-calibrated TOOL_REVIEW retry logic. Evaluation on the GAIA test set will establish robust cross-domain benchmarks. Investigation of multi-turn tool use via chained web retrieval and integration of the Tavily subsequent_request_token mechanism for follow-up queries represent natural next steps toward a fully grounded dynamic workflow agent.

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